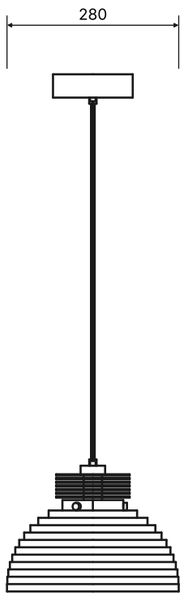
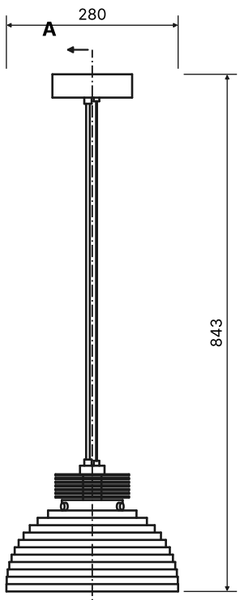
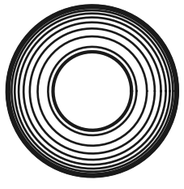
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

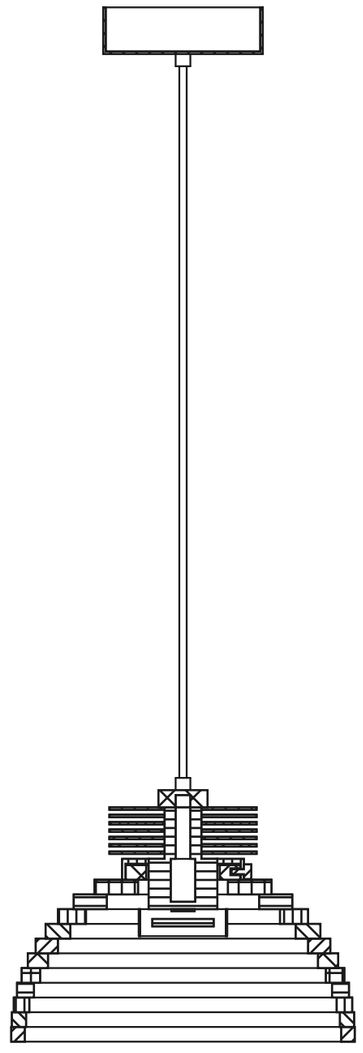
General arrangement

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. Section A-A (sheet 1): the product cut at x 0.0 and seen from the right; cut faces hatched, a different angle for each part.



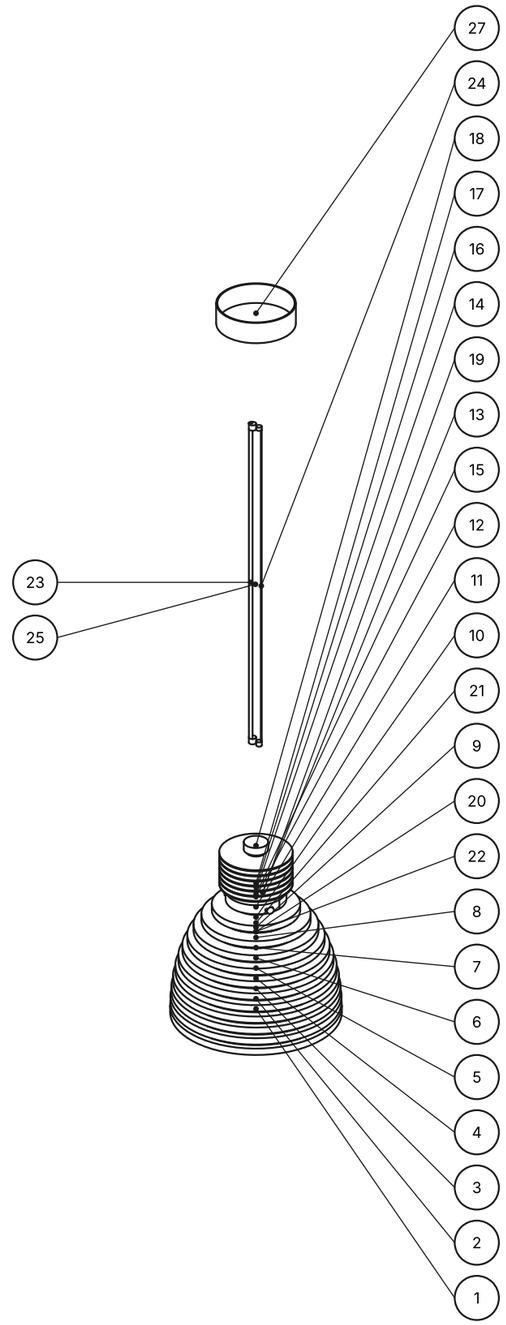
Section A-A

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).



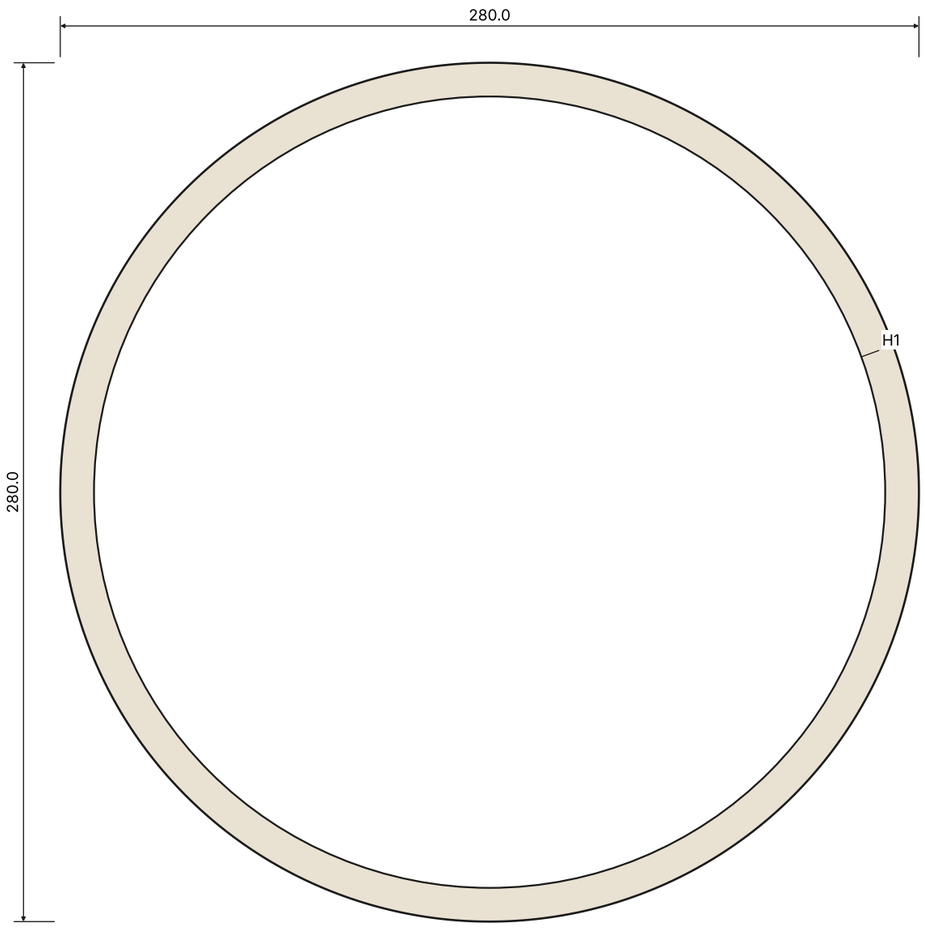
Exploded view

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/ring-01.dxf. Marks listed at the right.



ring-01: 12 mm cast-acrylic, 1 for the set, seen from its outer face

mark	size	x	y	depth
H1	Ø258.0	140.0	140.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: ring-01

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

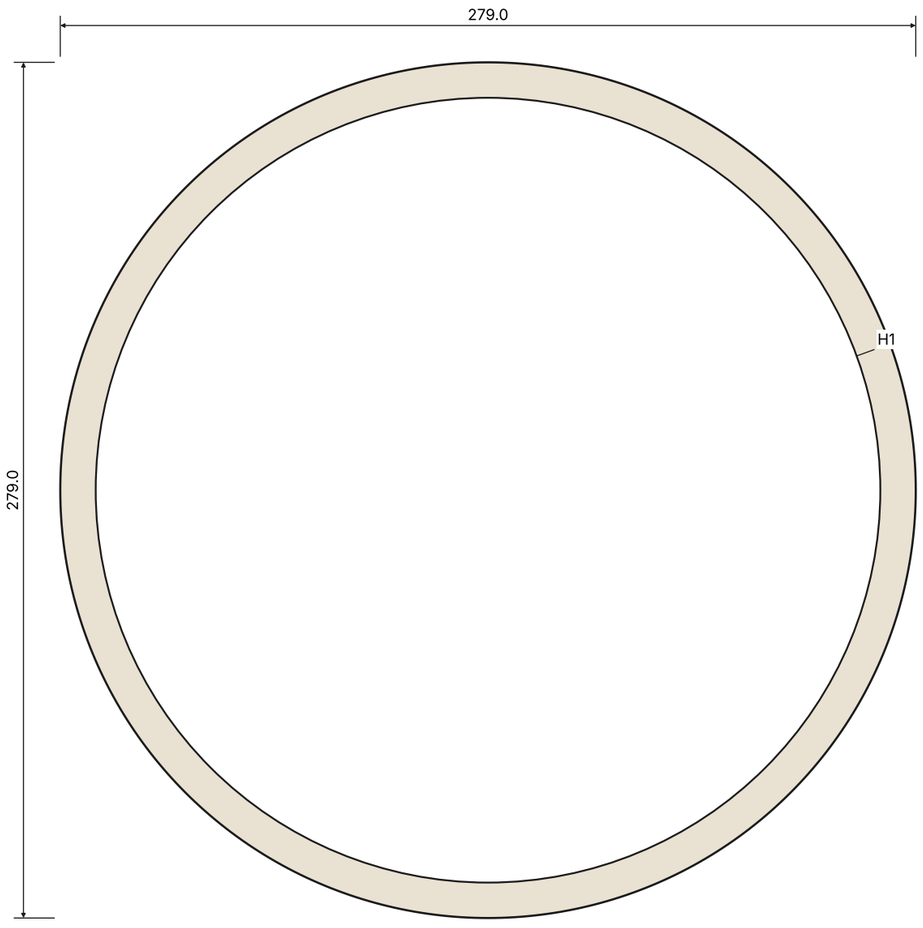
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/ring-02.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø255.9	139.5	139.5	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



ring-02: 12 mm cast-acrylic, 1 for the set, seen from its outer face

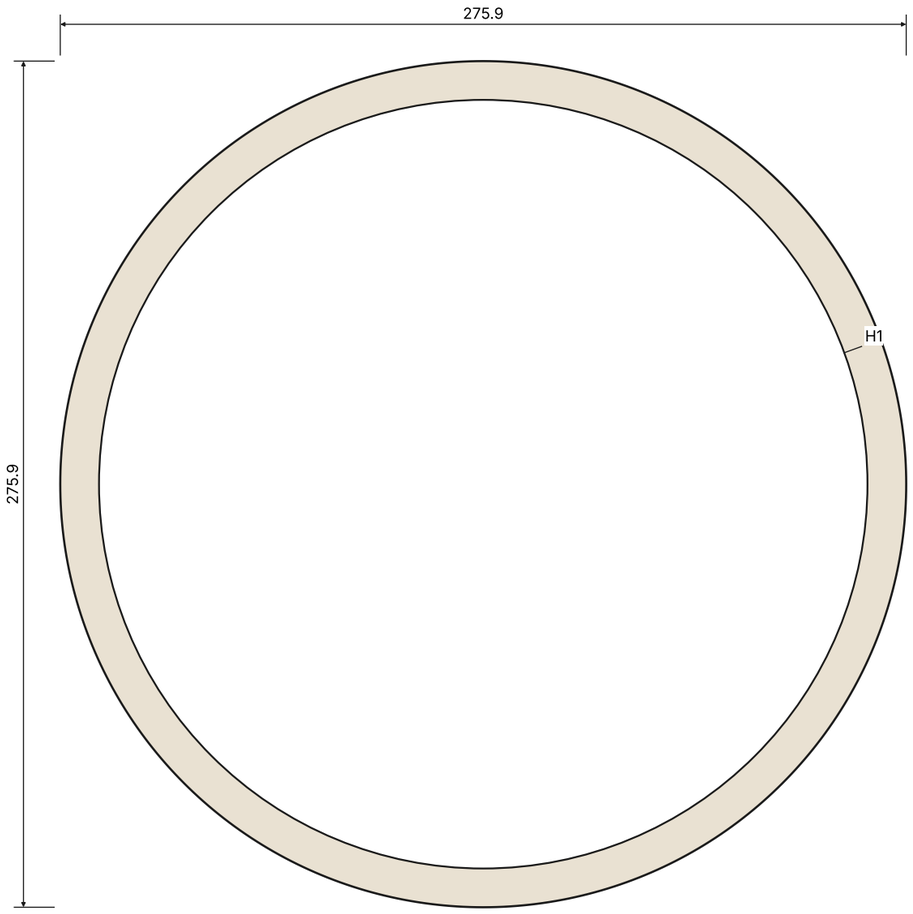
Sheet part: ring-02

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/ring-03.dxf. Marks listed at the right.



ring-03: 12 mm cast-acrylic, 1 for the set, seen from its outer face

mark	size	x	y	depth
H1	Ø250.6	137.9	137.9	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: ring-03

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

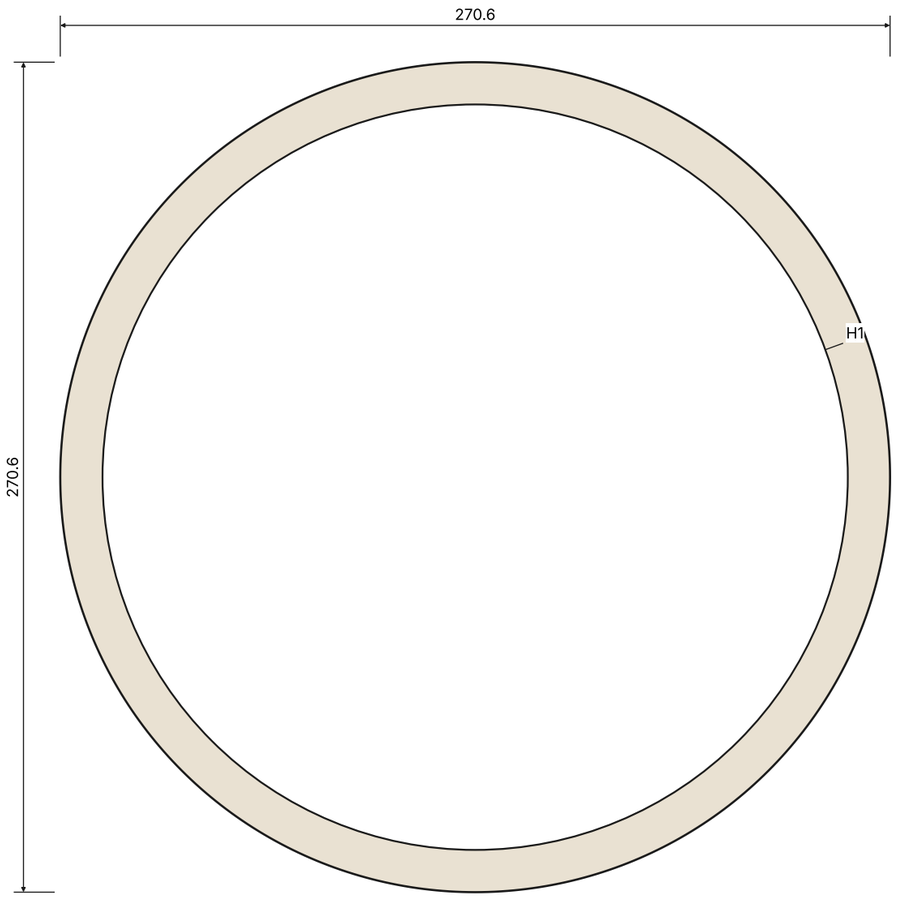
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/ring-04.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø243.0	135.3	135.3	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



ring-04: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: ring-04

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

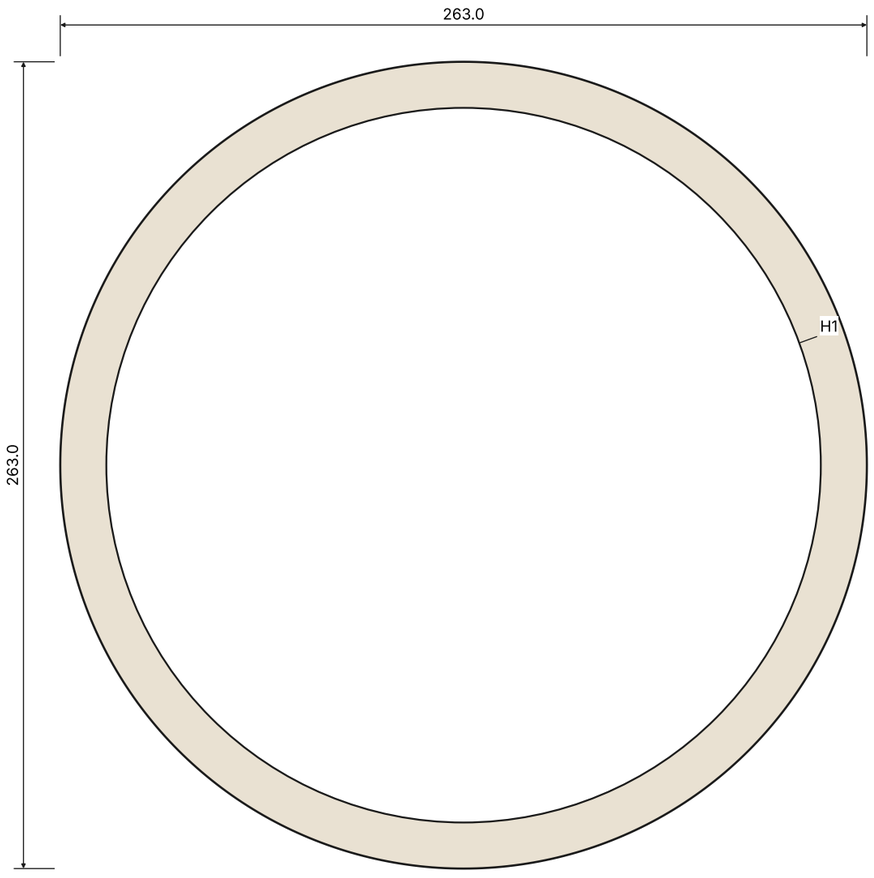
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/ring-05.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø233.0	131.5	131.5	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



ring-05: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: ring-05

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

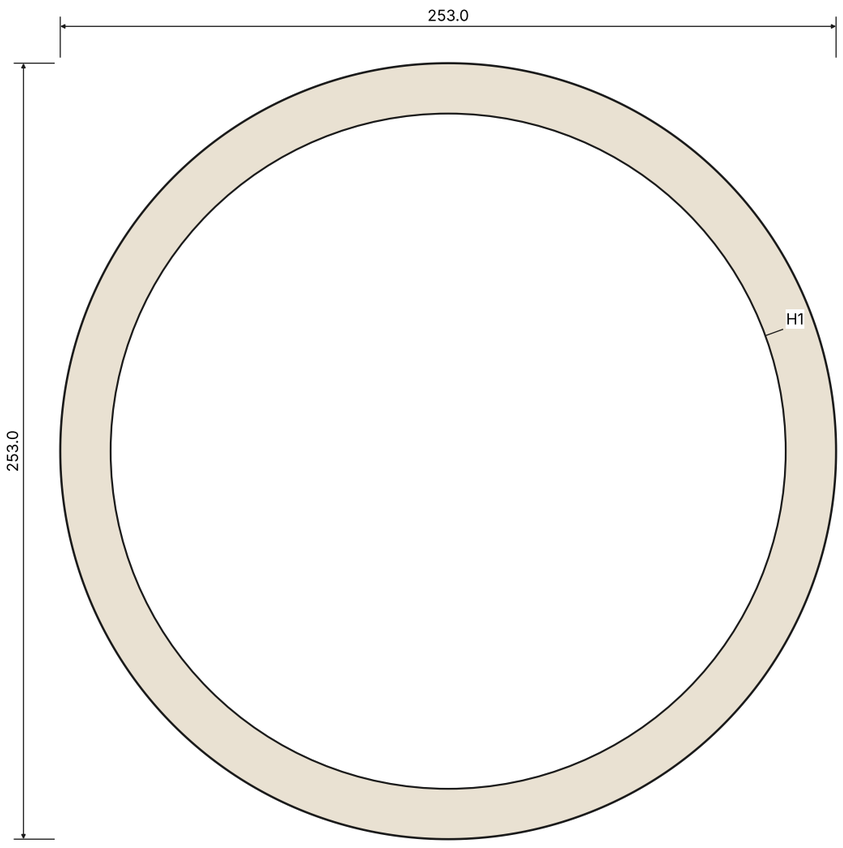
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/ring-06.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø220.1	126.5	126.5	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



ring-06: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: ring-06

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

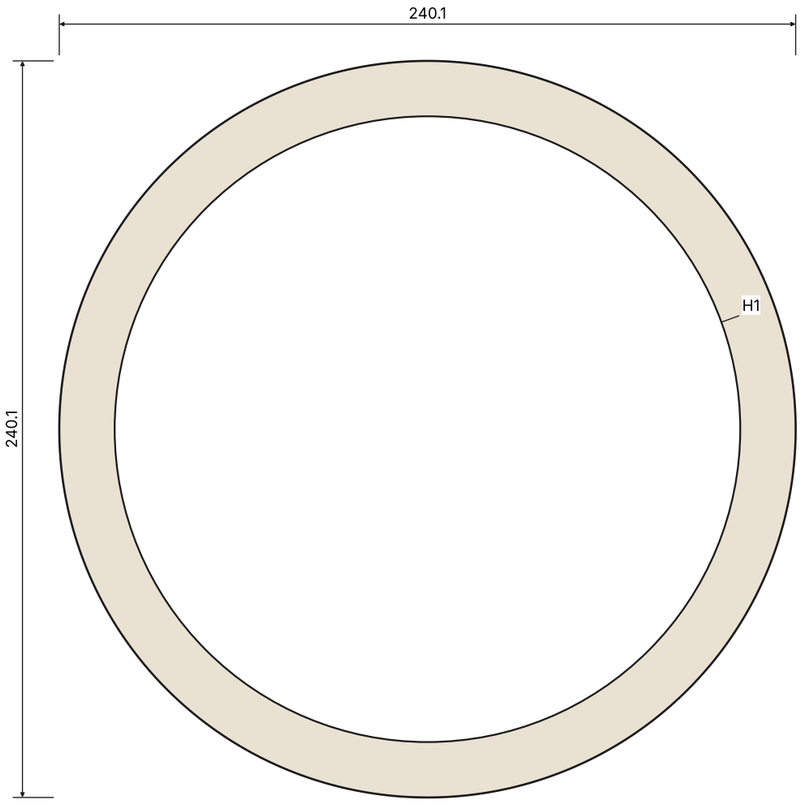
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/ring-07.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø204.0	120.1	120.1	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



ring-07: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: ring-07

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

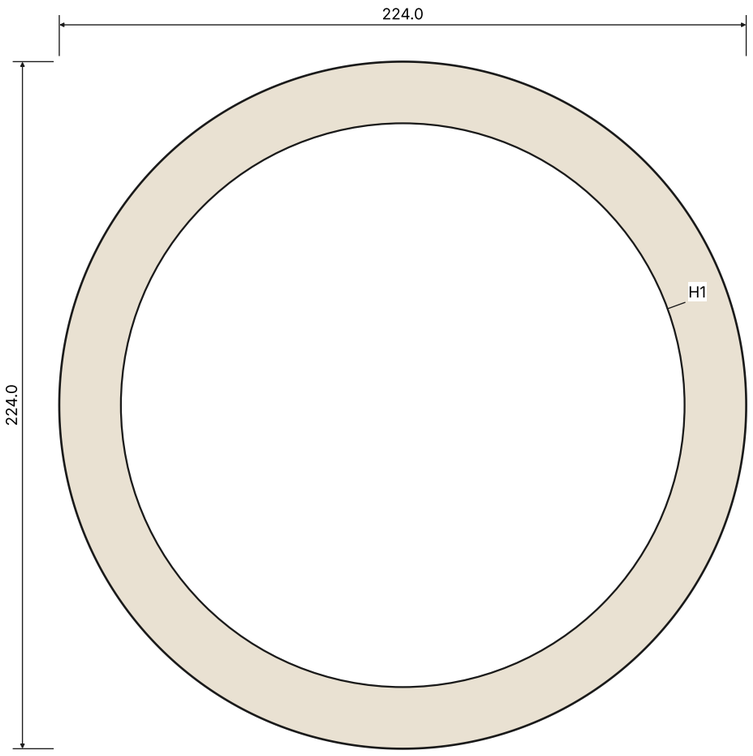
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/ring-08.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø183.8	112.0	112.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



ring-08: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: ring-08

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

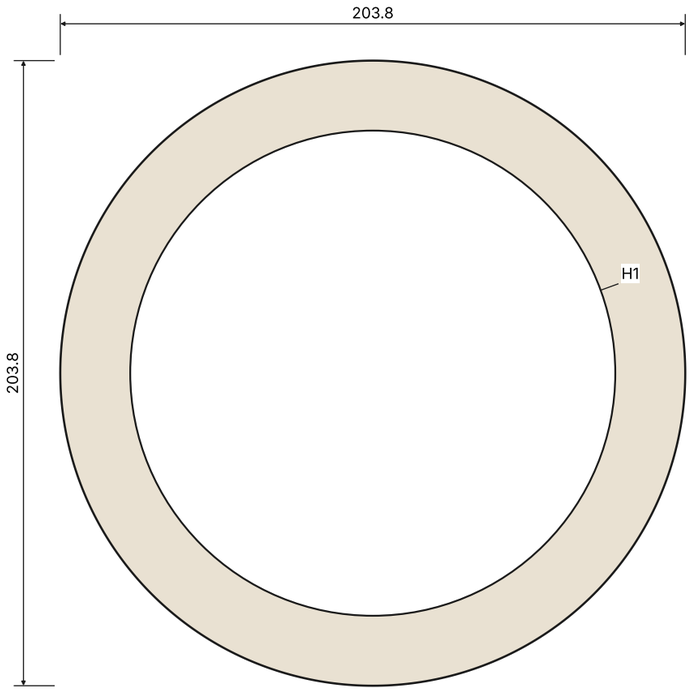
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/ring-09.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø158.2	101.9	101.9	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



ring-09: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: ring-09

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

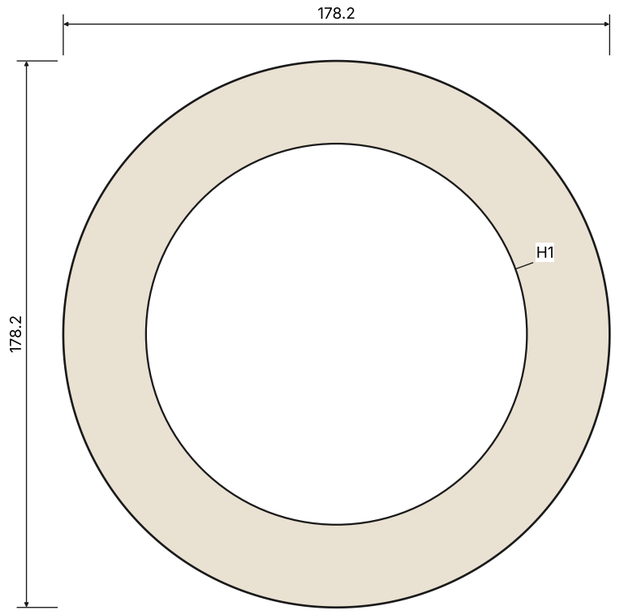
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/ring-10.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø124.2	89.1	89.1	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



ring-10: 12 mm cast-acrylic, 1 for the set, seen from its outer face

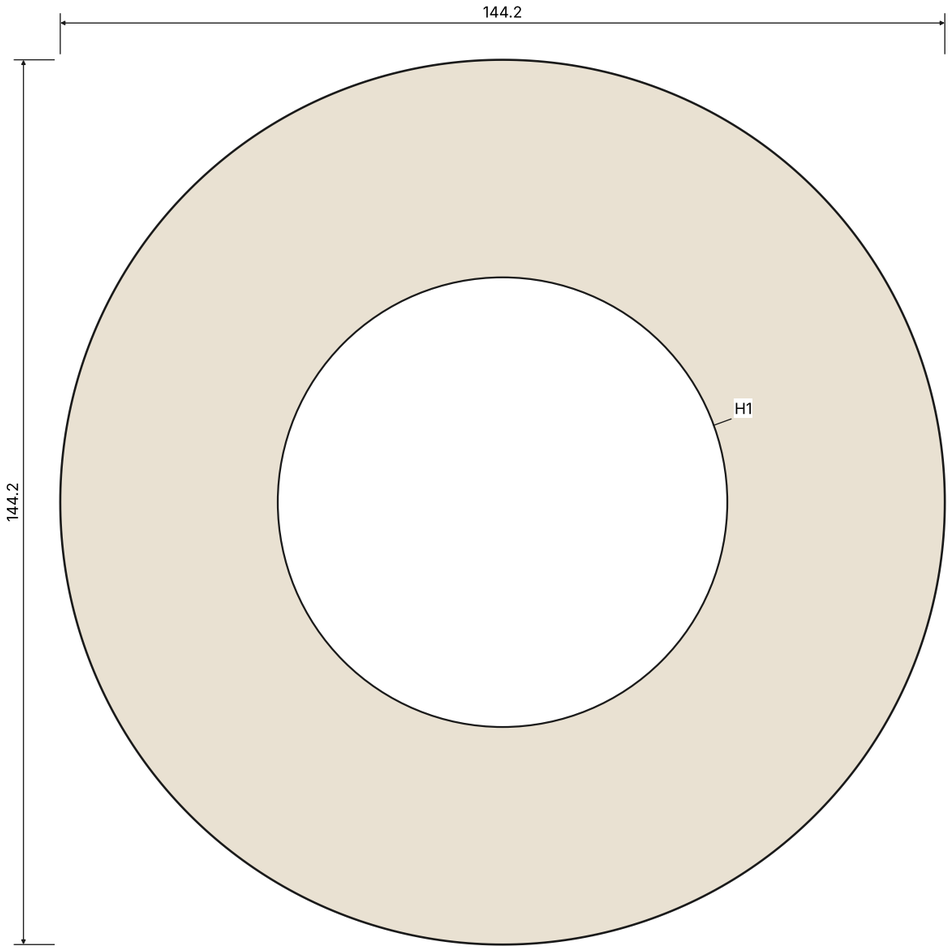
Sheet part: ring-10

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/ring-11.dxf. Marks listed at the right.



ring-11: 12 mm cast-acrylic, 1 for the set, seen from its outer face

mark	size	x	y	depth
H1	Ø73.3	72.1	72.1	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: ring-11

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

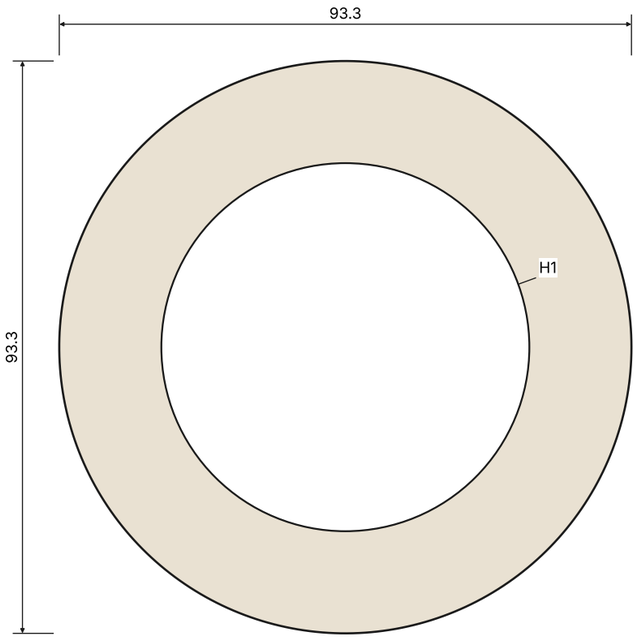
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/ring-12.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø60.0	46.6	46.6	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



ring-12: 12 mm cast-acrylic, 1 for the set, seen from its outer face

Sheet part: ring-12

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

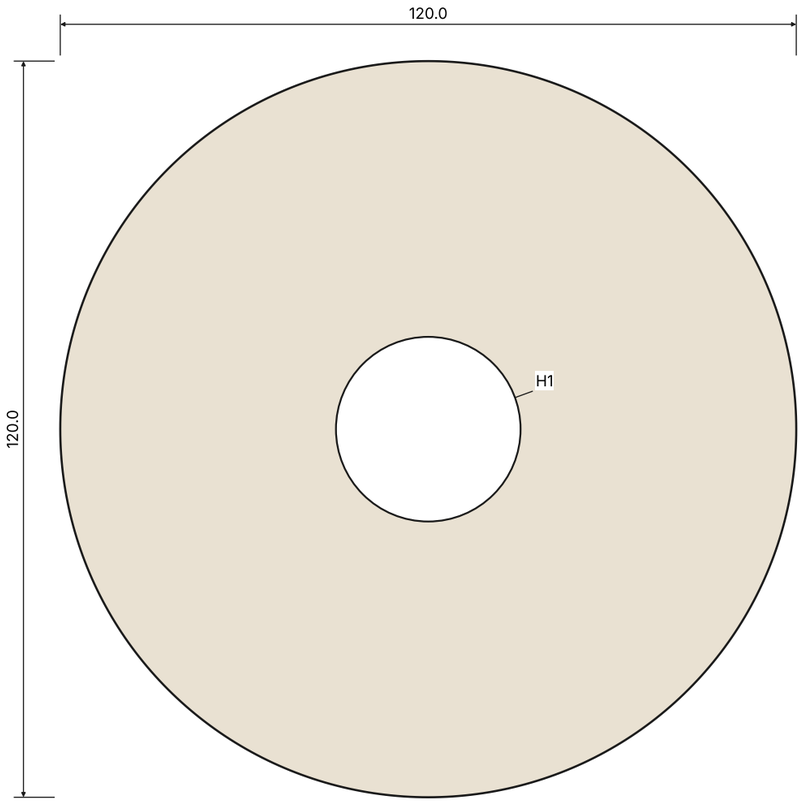
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/fin-1.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø30.1	60.0	60.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



fin-1: 2 mm aluminium, 1 for the set, seen from its outer face

Sheet part: fin-1

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

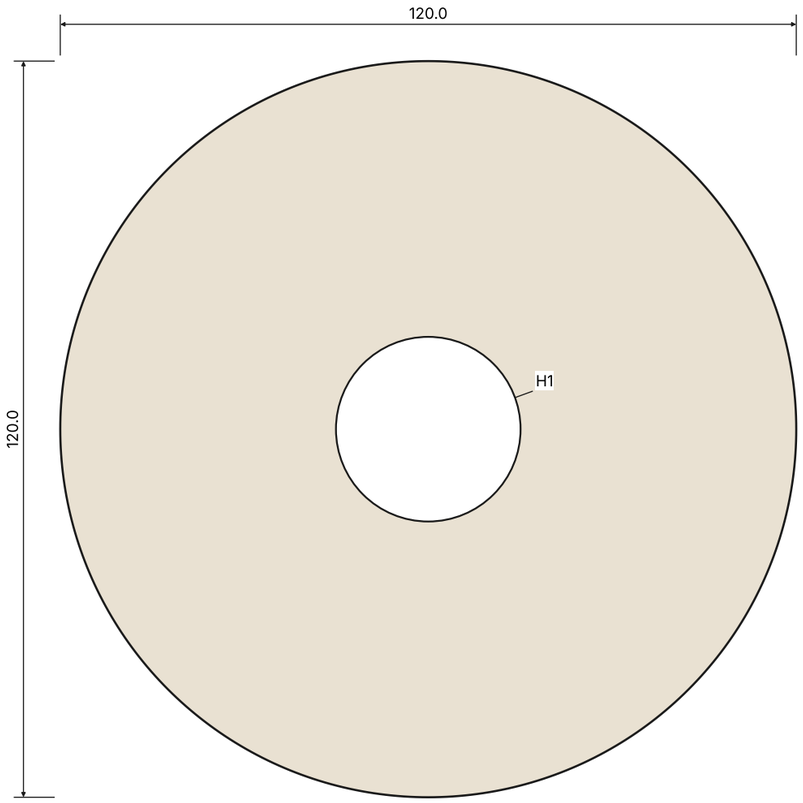
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/fin-2.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø30.1	60.0	60.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



fin-2: 2 mm aluminium, 1 for the set, seen from its outer face

Sheet part: fin-2

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

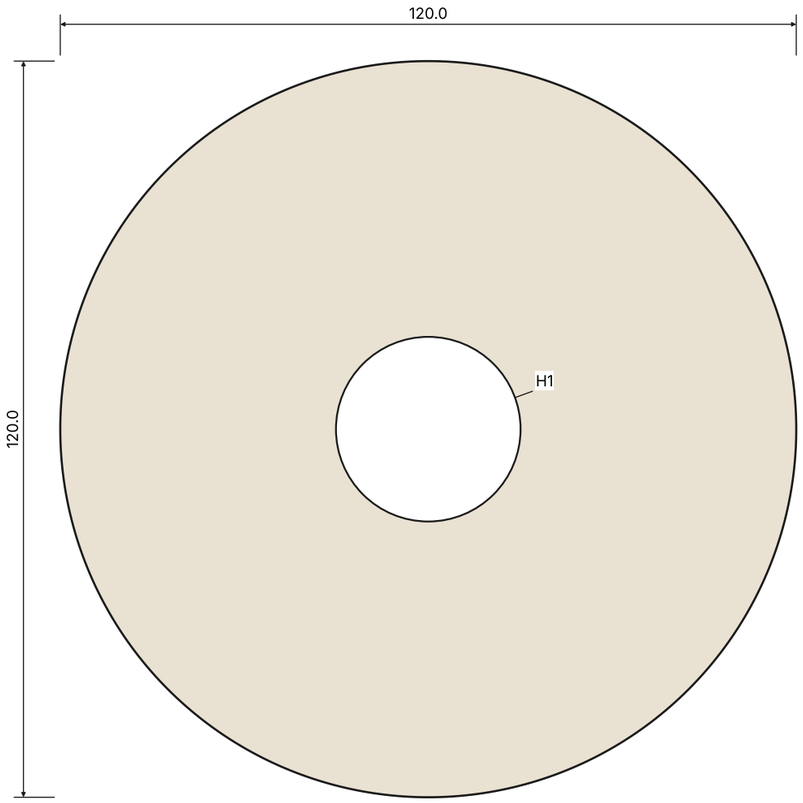
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/fin-3.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø30.1	60.0	60.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



fin-3: 2 mm aluminium, 1 for the set, seen from its outer face

Sheet part: fin-3

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

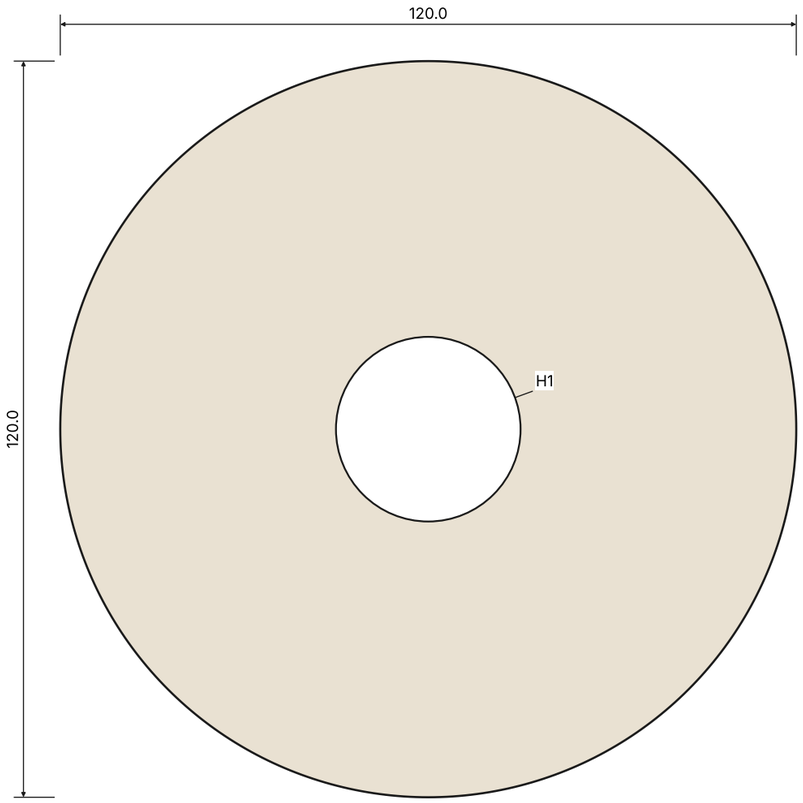
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/fin-4.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø30.1	60.0	60.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



fin-4: 2 mm aluminium, 1 for the set, seen from its outer face

Sheet part: fin-4

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

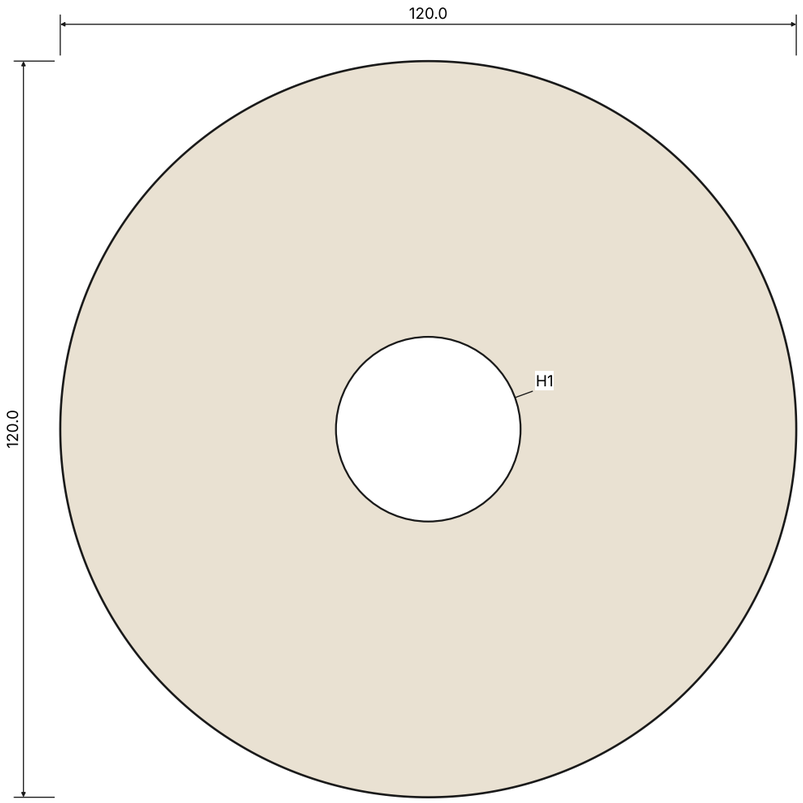
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/fin-5.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø30.1	60.0	60.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



fin-5: 2 mm aluminium, 1 for the set, seen from its outer face

Sheet part: fin-5

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

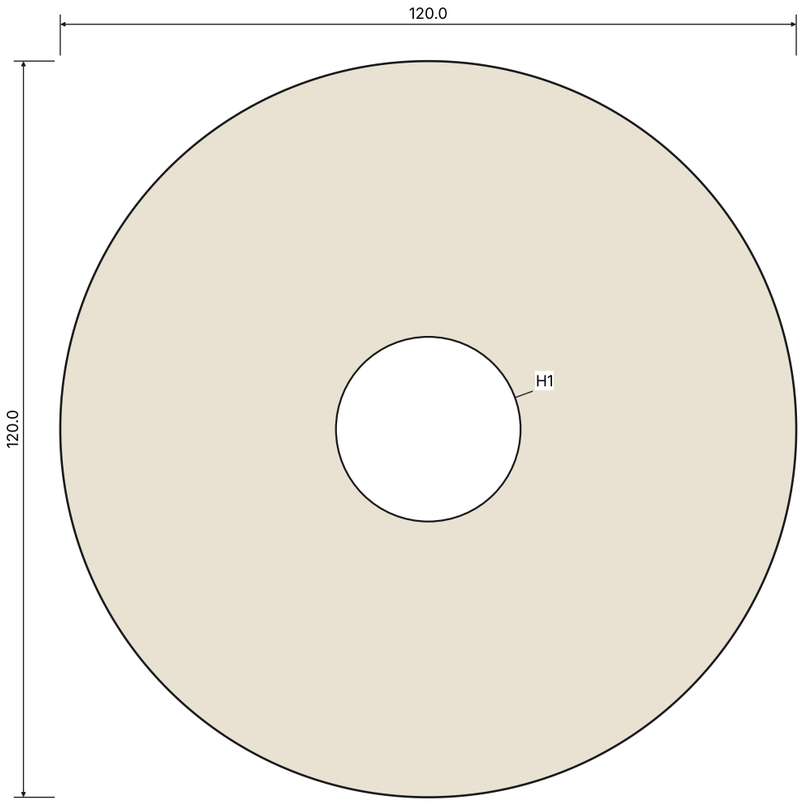
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/fin-6.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø30.1	60.0	60.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



fin-6: 2 mm aluminium, 1 for the set, seen from its outer face

Sheet part: fin-6

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

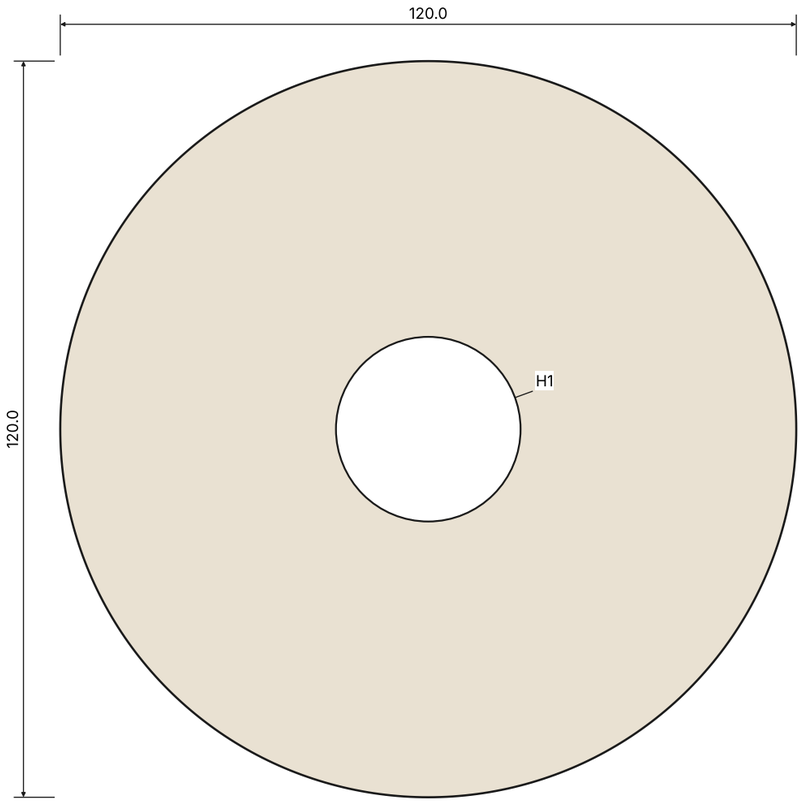
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/fin-7.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø30.1	60.0	60.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



fin-7: 2 mm aluminium, 1 for the set, seen from its outer face

Sheet part: fin-7

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

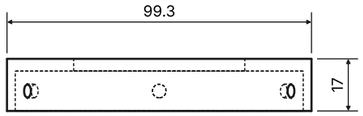
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

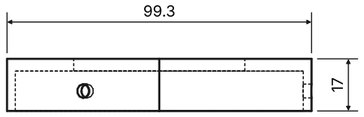
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

15. flange: lathe, aluminium; 32 cm3 each; 1 for the set; 1:2

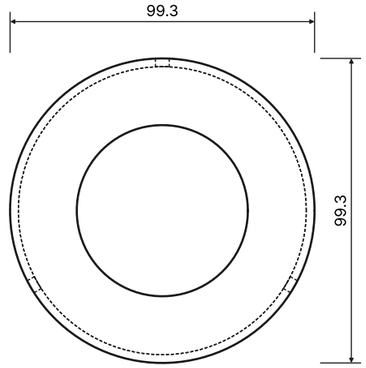
6061, turned, bead-blasted and black anodised: a 99 disc with a skirt down over the top ring's edge, three 4.5 holes radially for the thumbscrews



FRONT



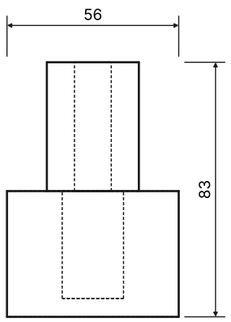
FROM THE RIGHT



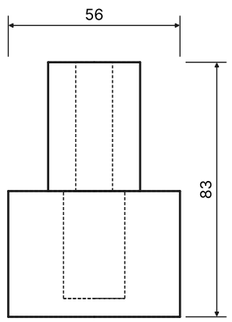
FROM ABOVE

16. core: lathe, aluminium; 115 cm3 each; 1 for the set; 1:2

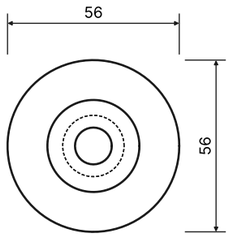
6061 from 2 1/2 in bar: the COB's seat (an Ideal Chip-Lok holder, two M3) faced flat on its underside, thermal pad; up through the top ring with 2 mm clearance; its spigot above takes the fins on spacers and the cap, an M20 thread



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

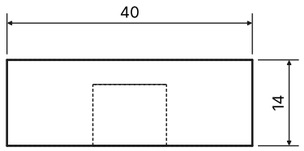
mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

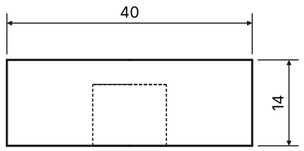
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

18. cap: lathe, aluminium; 16 cm3 each; 1 for the set; 1:1

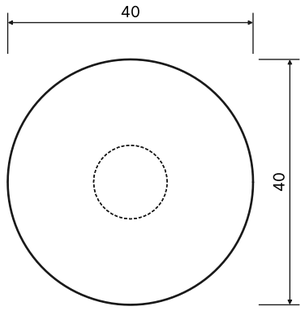
the cord grip and the cable gripper go into its top, side by side



FRONT



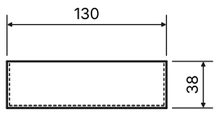
FROM THE RIGHT



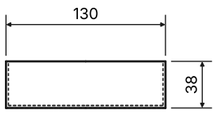
FROM ABOVE

27. canopy: lathe, aluminium; 55 cm3 each; 1 for the set; 1:5

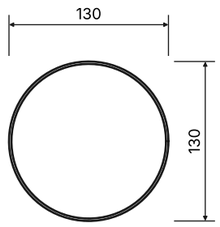
spun or turned aluminium, black anodised; the ERP driver inside, on the crossbar



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	ring-01	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	280 × 280	22.34
2	ring-02	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	279 × 279	22.18
3	ring-03	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	276 × 276	21.69
4	ring-04	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	271 × 271	20.87
5	ring-05	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	263 × 263	19.72
6	ring-06	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	253 × 253	18.24
7	ring-07	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	240 × 240	16.43
8	ring-08	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	224 × 224	14.30
9	ring-09	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	204 × 204	11.84
10	ring-10	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	178 × 178	9.05
11	ring-11	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	144 × 144	5.93
12	ring-12	cut	1	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	93 × 93	2.48
13	inserts	buy	3	any (E-Z LOK, Tappex or alike) Brass threaded insert for plastics, M4, heat-set, 5.6 OD x 8 long		0.50
14	gasket	buy	1	1 mm silicone sheet ring under the sink's flange, cut on a cutter plotter		1.00
15	flange	lathe	1	Aluminium 6061	109 × 109 × 27	53.06
16	core	lathe	1	Aluminium 6061	66 × 66 × 93	55.13
17	fin-1, fin-2, fin-3, fin-4, fin-5, fin-6, fin-7	cut	7	2 mm Aluminium 5052 sheet	120 × 120	2.88
18	cap	lathe	1	Aluminium 6061	50 × 50 × 24	46.50
19	thumbscrews	buy	3	any (a machine shop, or Misumi knurled thumb screws) M4 knurled thumb screws, 304 stainless, 12 head x 6, shoulder 8, captive in the sink's flange		4.00
20	reflector	buy	1	a machine shop The COB's reflector: a spun or turned aluminium cone, bead-blasted satin inside (so about a quarter of the light spills wide onto the rings), black outside		9.00
21	cob	buy	1	Bridgelux Vero SE 10 COB, 2700 K, CRI 90, at 180 mA about 590 lm, 24.6 V, 4.4 W (the 575 lm level), in an Ideal Chip-Lok holder		12.00
22	louvre	buy	1	any (a lighting louvre maker) Black anodised aluminium honeycomb louvre, 50 across, 3.2 cell, 6 deep: no direct view of the COB beyond about 30 deg off the axis		8.00
23	cord	buy	1	any UL listed 18/2 SVT cord, black rayon or cotton braid, 3 m cut to the drop (beside the suspension cable)		9.00
24	cable	buy	1	1.5 mm 7×7 stainless aircraft cable, 2 m, cut to the drop		3.00
25	grips	buy	2	Heyco (or alike, UL recognised) Black nylon liquid-tight cord grip, PG7 / 1/4 NPT class, for a round 18/2 cord, two (canopy and sink)		3.00
26	grippers	buy	2	any listed (Gripple, Bettercable or alike) Pendant cable gripper for 1.5 mm stainless aircraft cable, adjustable, M10 × 1 or 1/8 IPS thread, rated well over 10 kg, two (canopy and sink)		12.00
27	canopy	lathe	1	Aluminium 6061	140 × 140 × 48	68.52

Parts list (1 of 2)

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
28	driver	buy	1	ERP Power ESS010W-0180-42, 180 mA constant current, phase and 0-10 V dimming, Class 2 (components.json north-driver-575)		16.00
29	crossbar	buy	1	Canopy crossbar for a 4 in outlet box, two 8-32 screws, a ground screw		3.00

Parts list (2 of 2)

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. A proposal (strata/README.md): Hemisphere built from stacked rings. The current Hemisphere (dome_pendant.py) stays the baseline.
2. The shade: 12 rings of 1/2 in cast sheet, 280 across and 144 tall (152 at 12.7), each 11 to 35 wide, lapping the ring above by 10 (the bond). Routed from sheet with no mould; bonded in a stepped jig with polymerising cement; annealed; the outer edges diamond-polished, the inner edges bead-blasted. About 1.88 kg of acrylic.
3. The light: a Bridgelux Vero SE 10 at 180 mA (about 590 lm, 4.4 W), recessed 36 up the shade in a satin reflector, behind a 3.2 honeycomb louvre: no view of the COB beyond about 30 deg off the axis. A quarter of its light spills from the satin reflector onto the rings' blasted inner edges, and leaves through their polished outer edges: each terrace glows as a line.
4. Heat (worked): about 3.3 W into seven 120 fins (0.154 m2 of black anodise), at a low h for their close pitch: about 6 K over the room air. The COB's case runs about 40 C, and the top ring under its gasket well under 70 C. Against the current top (about 0.010 m2, about 31 K), about ten times the area.
5. Weight: about 2.5 kg on a 1.5 mm stainless cable (break about 1.7 kN) and a listed gripper; the cord carries current only (certification.md: the shade hangs from a support, not the cord).
6. Service: three thumbscrews and the shade lifts off the engine; the COB is on two screws in a Chip-Lok holder; the driver is in the canopy. Every part can be replaced.

Notes

Terrace: Pendant: a dome of twelve stacked tint rings, its terraces lit

mm | issue 2026-10-31 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15